

PORTFOLIO · 2026

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I take ideas to shipped software, alone, with AI agents — and set the rules that keep them from breaking.

UBC Sauder, Class of 2026 · Finance + Business Analytics

Open to anywhere in Canada

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SIX MONTHS, ONE PERSON

Four kinds of things I built

The numbers are output, not users.

38

projects

Delivered to others

- Reference-letter system
in use by a professor
- SQL teaching explorer
no-code tool for a professor

801

logged updates

Public

- Scholar Outflow Lab
13 countries, updates daily
- Earnings-call sentiment
tone → residual signal
- Ghost-job trading signal
310K postings

25

apps on one private database

Used every day

- 25-app personal operating
system

8

public repos

Runs on its own

- Multi-agent content pipeline
topic to release pack
- 8 scheduled jobs
- Guardrails and sentinels
machine blocks dangerous
operations

FOUR WORKS

What I built, and the decision inside each

WORK 1 · DELIVERED TO A PROFESSOR

Reference-letter system

Roster in, students screened by grade and attendance, documents uploaded, emails out. Shipped as a web app in a day; rebuilt as a desktop app after reading the privacy law, so student data stays on the professor's machine.

3 months · 55 iterations · Python / Flask / SQLite / OCR

THE POINT Nobody asked. I read the law and changed the architecture.

WORK 3 · RUNS ON ITS OWN

Multi-agent content pipeline

Topic to release pack across several content channels. Agents exchange files, never chat; resumable state machine, bounded retries, a QC loop that sends failures back, and a human always publishes.

32 modules · 20 skills · usage ledger reconciled to the platform bill

THE POINT My own ledger covered 1% of the real bill. The bill became the source of truth.

WORK 2 · USED EVERY DAY

A 25-app personal operating system

Single-file apps with no build step and one shared convention set, all on one private database. The website reads only data marked public and has zero access to the rest.

25 apps · 10 with built-in AI · count gate before every deploy

THE POINT Permission boundaries fixed in the architecture, not in habits.

WORK 4 · KNOWLEDGE

Structured knowledge bases built with AI

Thousands of entries across several private libraries. AI does the entry work; every entry carries a permanent ID and a source; reuse is by pointer, never by copy.

Verbatim first · no lossy simplification · checks before and after every change

THE POINT The hard part is the acceptance discipline, not the AI.

Six public projects

Each one is a complete, documented tool.

SQL-Viz-for-Edu

No-code SQL teaching tool for a professor: click tables, filters and groups, the SQL appears live. Deterministic by design, no LLM.

Single-file HTML + vanilla JS
[NickkkLian/SQL-Viz-for-Edu](https://github.com/NickkkLian/SQL-Viz-for-Edu)

Scholar-Outflow-Lab

Where researchers from 13 countries end up, from OpenAlex open data. Ranked by Wilson lower bound so small samples sink. Updates daily.

Python stdlib scraper · GitHub Pages · launchd
[NickkkLian/Scholar-Outflow-Lab](https://github.com/NickkkLian/Scholar-Outflow-Lab)

Earnings-Call Sentiment

Management framing vs analyst pushback, by topic, across quarters. Does tone explain the residual move net of the EPS surprise?

Python · tool use + Pydantic · React dashboard
[NickkkLian/Earnings-Call-Sentiment-Analyser](https://github.com/NickkkLian/Earnings-Call-Sentiment-Analyser)

Ghost-Job Trading Signal

Ghost postings detected in 310K listings, turned into a cross-sectional signal. Walk-forward backtest; +3.7pp/yr vs SPY, 2019–2026. Team of three.

XGBoost · GMM · SHAP · Revelio data
[NickkkLian/Ghost-Job-Detection-And-Trading-Signal](https://github.com/NickkkLian/Ghost-Job-Detection-And-Trading-Signal)

content-organizer

Saved Xiaohongshu posts and Bilibili videos in one library; Claude merges selected notes into one writeup. Data stays in your browser and your own repo.

Single-file app · IndexedDB · MIT · live demo
[NickkkLian/content-organizer](https://github.com/NickkkLian/content-organizer)

Job-Tracker

A multi-region job-search tracker in one HTML file: prompt generators, batch resume tailoring, and the GitHub Contents API as its backend.

React 18 in one HTML file · GitHub Contents API
[NickkkLian/Job-Tracker](https://github.com/NickkkLian/Job-Tracker)

Ship first, then let machines keep it honest

Every rule comes from a real incident. I don't hand-write code: I define the problem, set the constraints, verify the result, and own the result.

Keeping systems reliable

- A pre-tool hook blocks dangerous operations
force push, mass deletion, pushing while behind · 332 lines · ask, never deny
- Sentinels raise alarms
database shrinks, copies drift, sessions collide
- The reviewer is never the doer
an audit agent judges evidence only
- Three backup tiers, restores actually rehearsed

What I can do

- Take an idea to production alone
- Set the rules that keep systems reliable
- Python backends, single-file frontends, agent orchestration
13–15 AI sessions in parallel, one repo each
- Product judgment: what is worth building, what to cut
- Business grounding: valuation, financials, backtesting
bank valuation work · CFA Level I